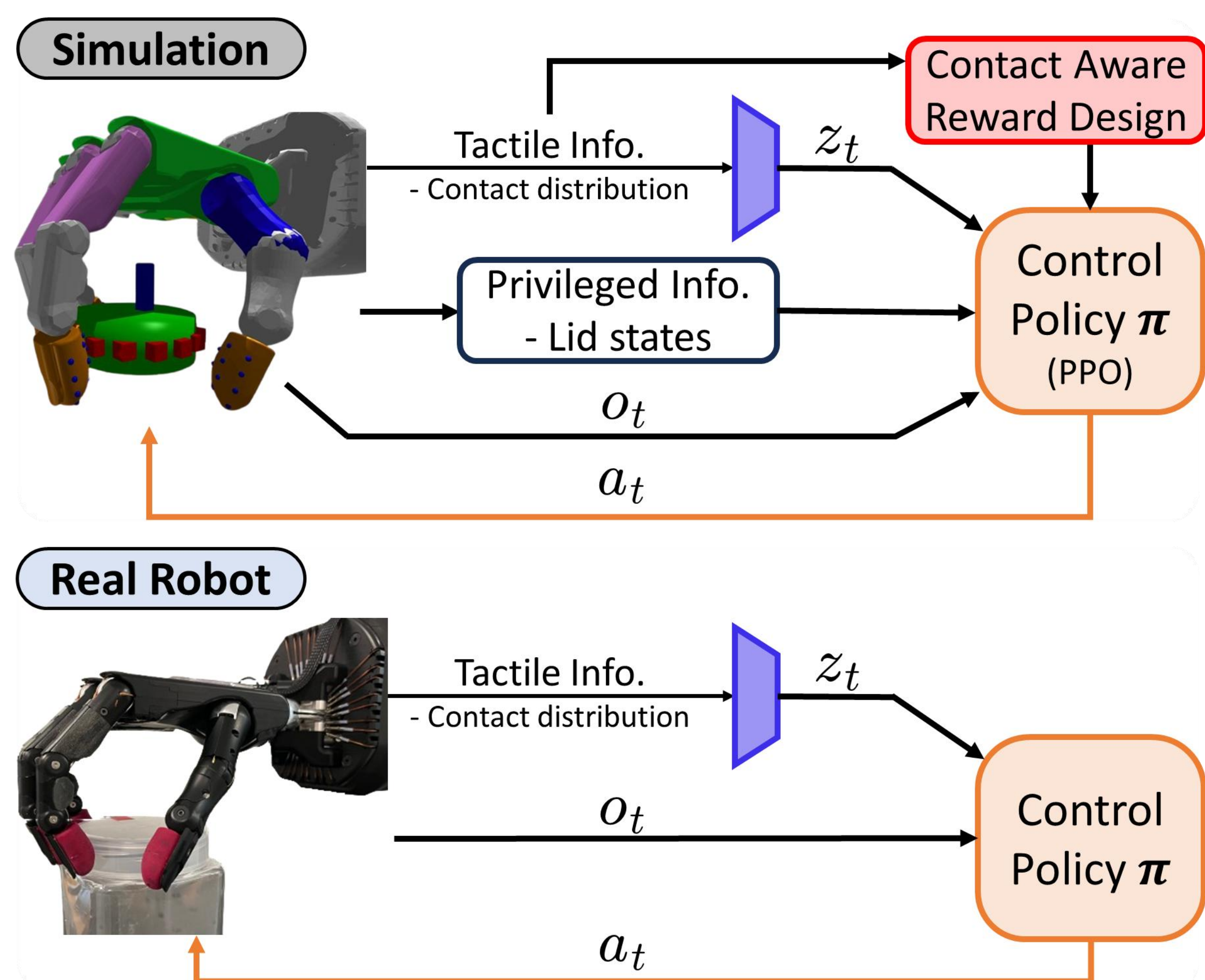


Tac2Motion: Contact-Aware Reinforcement Learning with Tactile Feedback for Robotic Hand Manipulation

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Introduction: In this work, we propose a new RL-based framework for in-hand manipulation in contact-rich environments such as lid-removing tasks. To this end, the main idea is **to incorporate tactile sensing into observation and reward functions to guide contact-aware control policy**. We approximate and simulate tactile sensing through multiple contact penetrations. Lastly, the proposed framework is demonstrated on removing a lid of the bottle, including a comparison to the baseline.

Structure of the framework



Contributions

- We utilize **tactile sensing** in the **observation space** to estimate the object dynamics such as torsional friction of a lid. This allows an agent to adapt to uncertain dynamics of the lid.
- We leverage the contact pressure captured by **tactile sensing** to shape the **reward function** that maximizes friction between the fingers and the lid. This shows a major departure from the existing use of tactile sensing modality in RL.
- We introduce a **virtual torque** on the lid induced by contact friction, which is generated from contact pressure. This eliminates the necessity of complex patch contact modeling and enables **efficient and stable training of removing the lid based on tactile-inspired dynamics**.

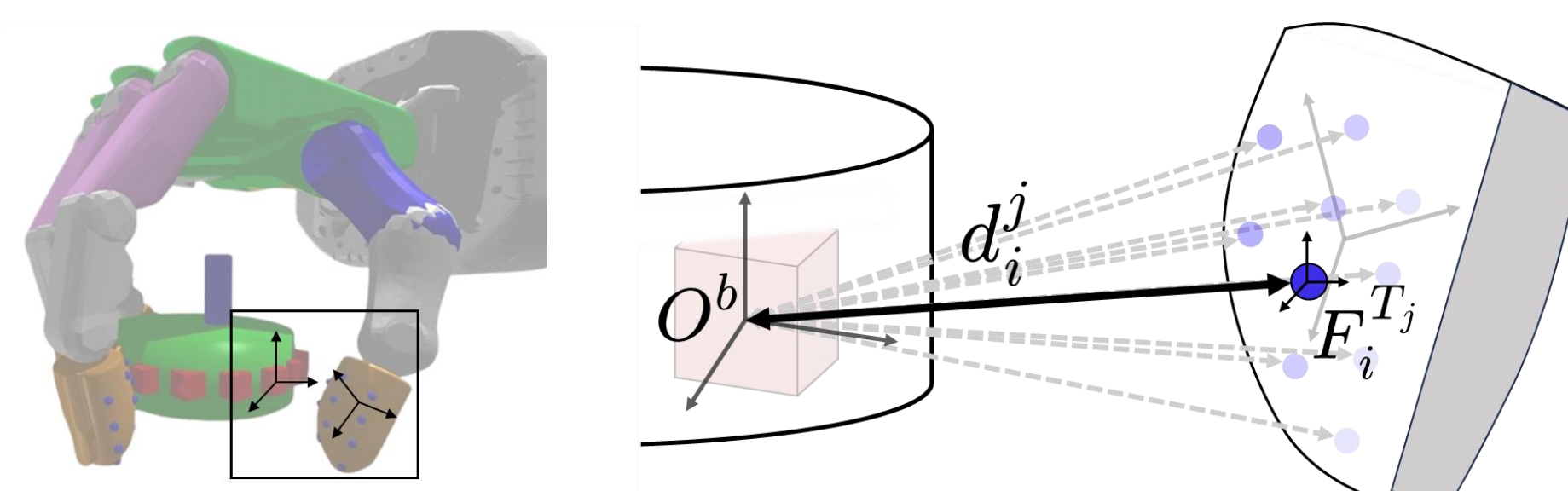
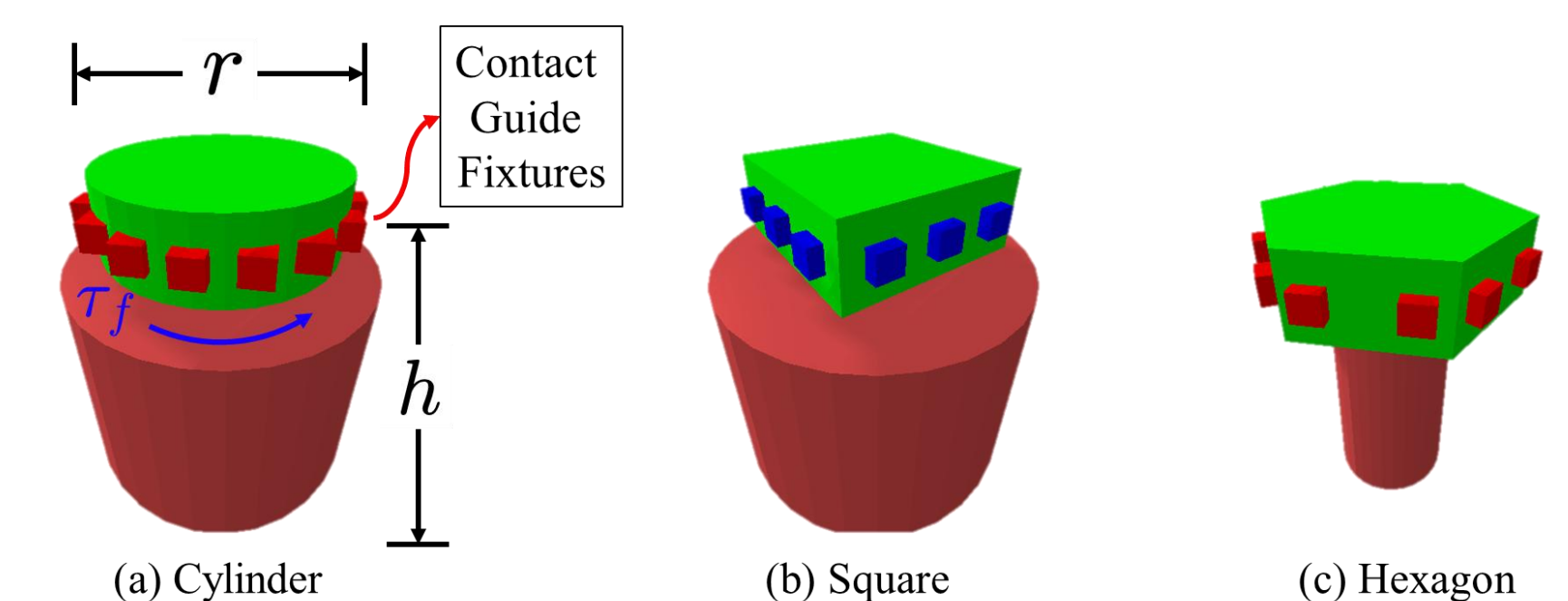
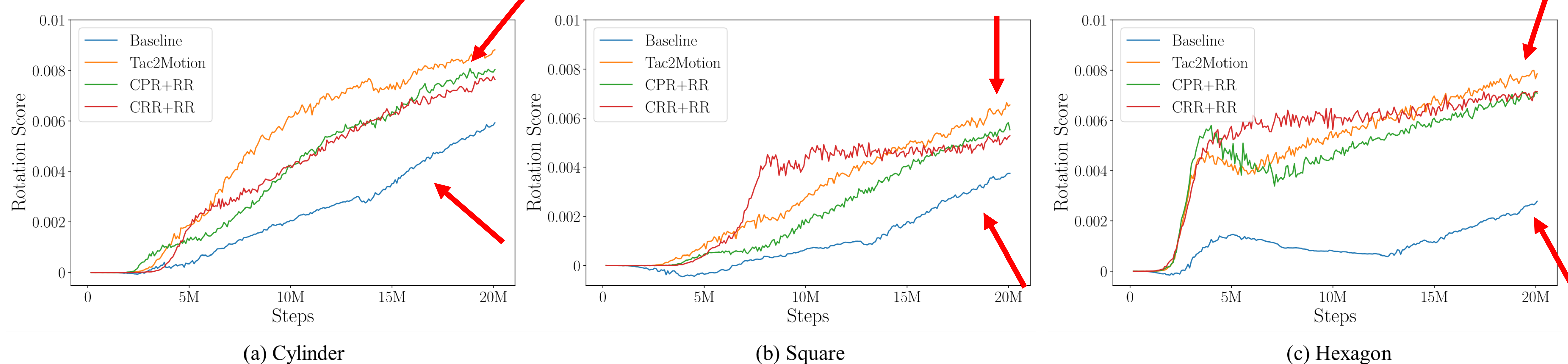


Illustration of calculation of contact pressure in the simulation



Types of the lid in the simulation

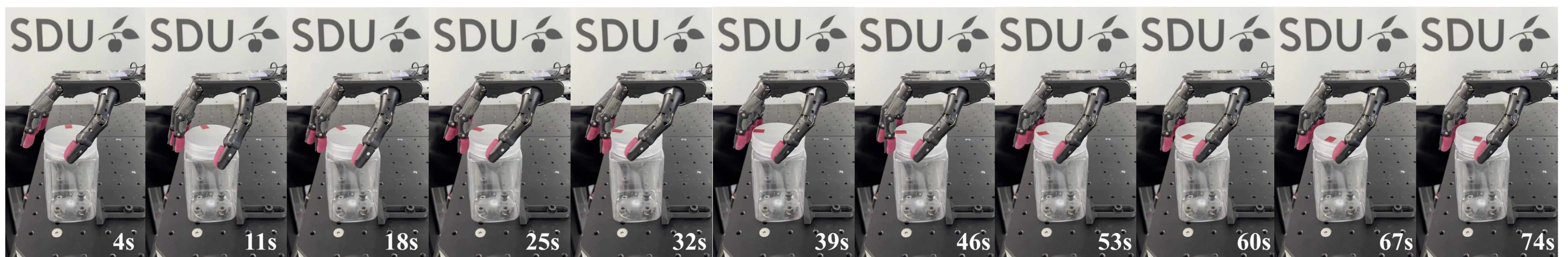
Simulation Results



Methods	Rotation Score (RS)			Average
	Cylinder	Square	Hexagon	
Baseline [15]	0.0087 ±0.0007	0.005 ±0.0005	0.0035 ±0.0002	0.0057
Tac2Motion	0.0108 ±0.0004	0.0086 ±0.0007	0.0094 ±0.0007	0.0096
CPR+RR	0.0097 ±0.0003	0.0078 ±0.0005	0.008 ±0.0009	0.0085
CRR+RR	0.009 ±0.0003	0.0068 ±0.0007	0.007 ±0.0005	0.0076

- Training performance of each method over 20M steps across different types, sizes, and object dynamics. The results show that **Tac2Motion** learns the torsional motion faster than other methods and achieves **superior final performance**, showing **improved data efficiency**.

Qualitative Experimental Results



- Snapshots from experimental deployment of the learned control policy for removing the lid with a multi-fingered robotic hand.

Conclusion

- We verify that **Tac2Motion outperforms the baseline** in opening a lid, observing **higher data efficiency** and **robust behaviors**. Lastly, we demonstrate **the learned control policy on a real robotic setup** and show that it can actually remove the lid smoothly.